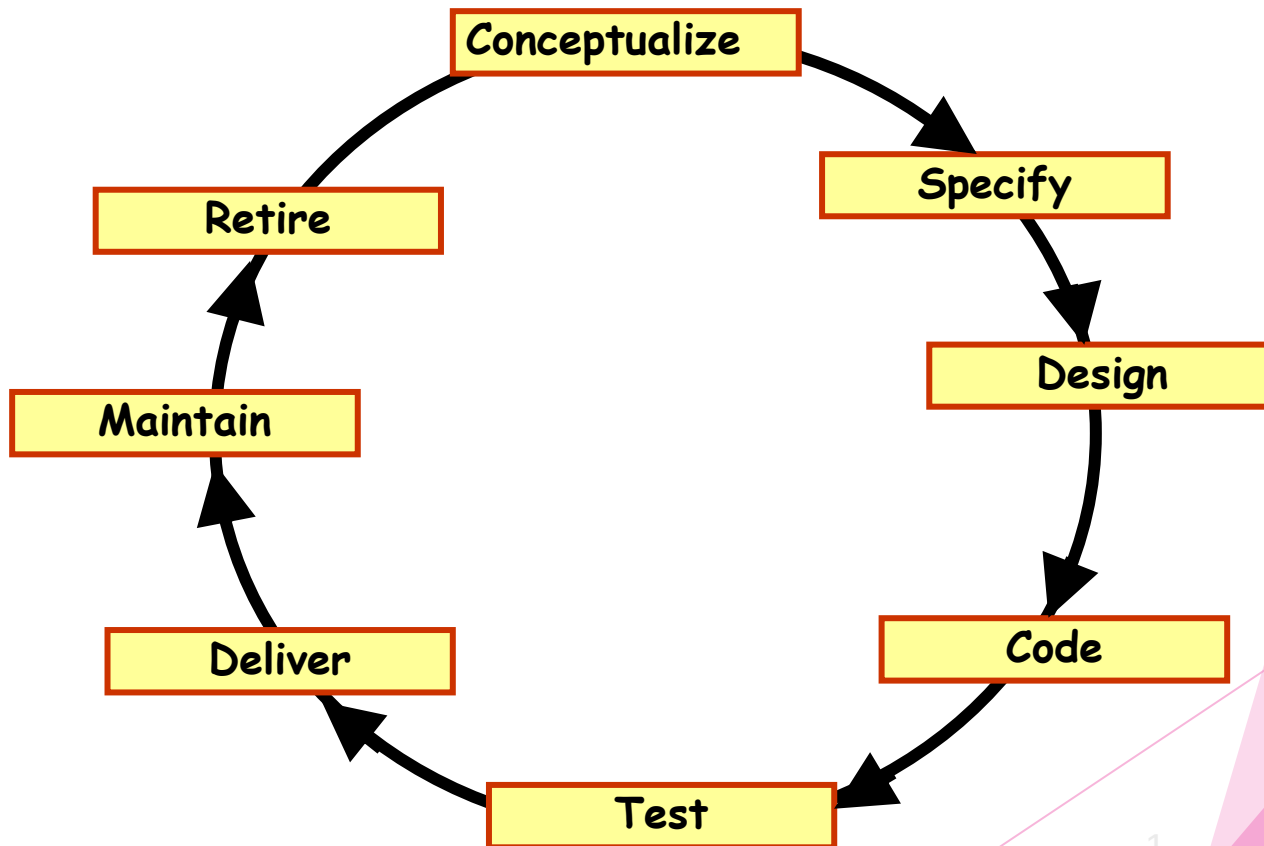
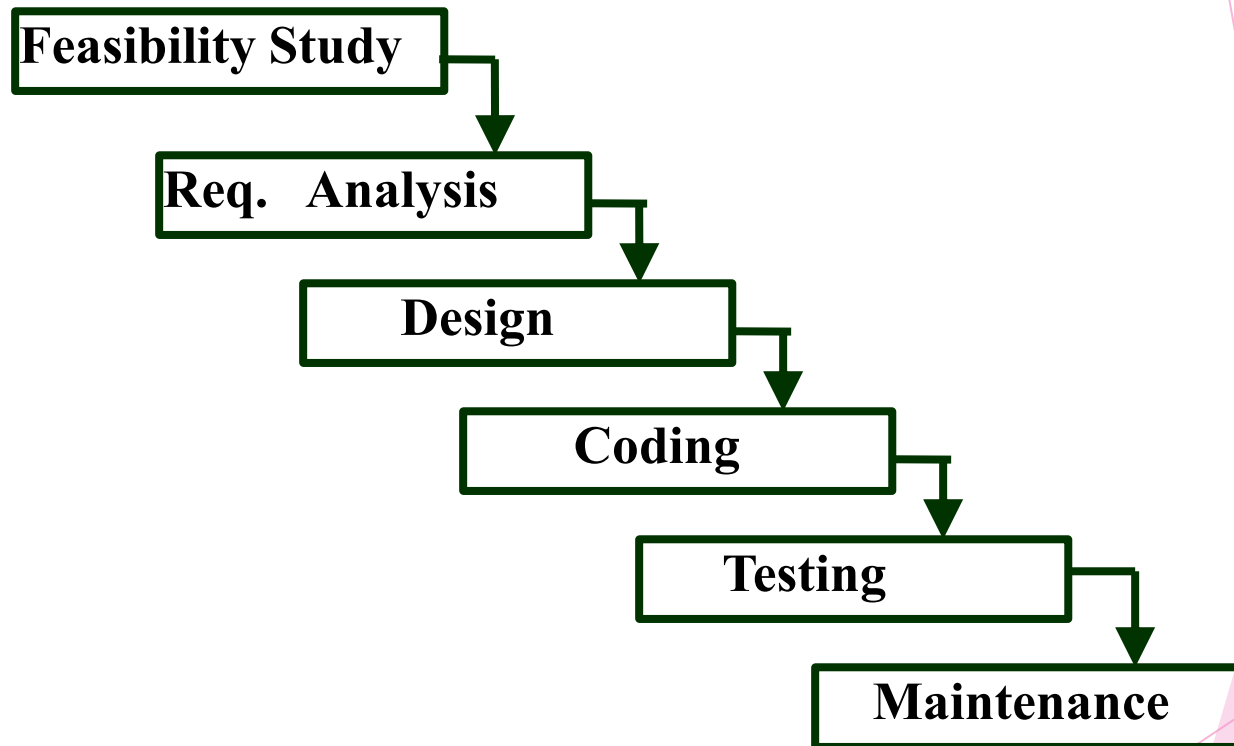


Software Life Cycles



Classical Waterfall Model



Classical Waterfall Model

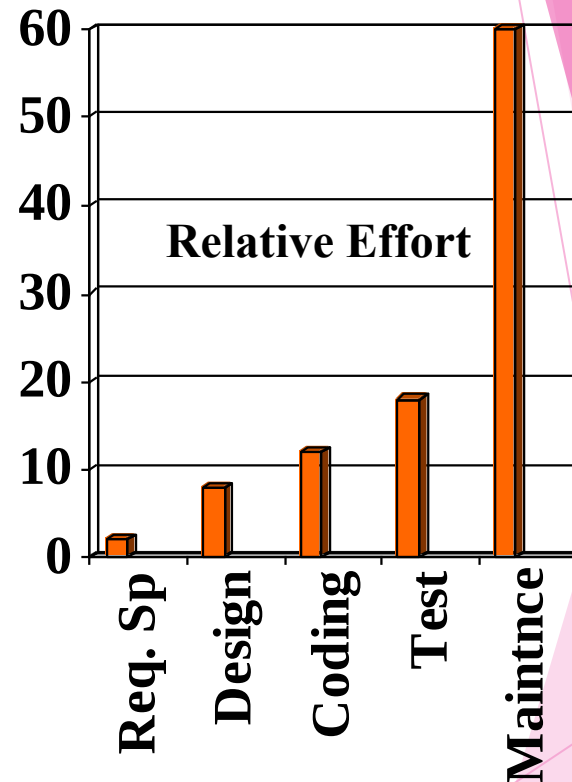
► Classical waterfall model divides life cycle into phases:

- feasibility study,
- requirements analysis and specification,
- design,
- coding and unit testing,
- integration and system testing,
- maintenance.



Relative Effort for Phases

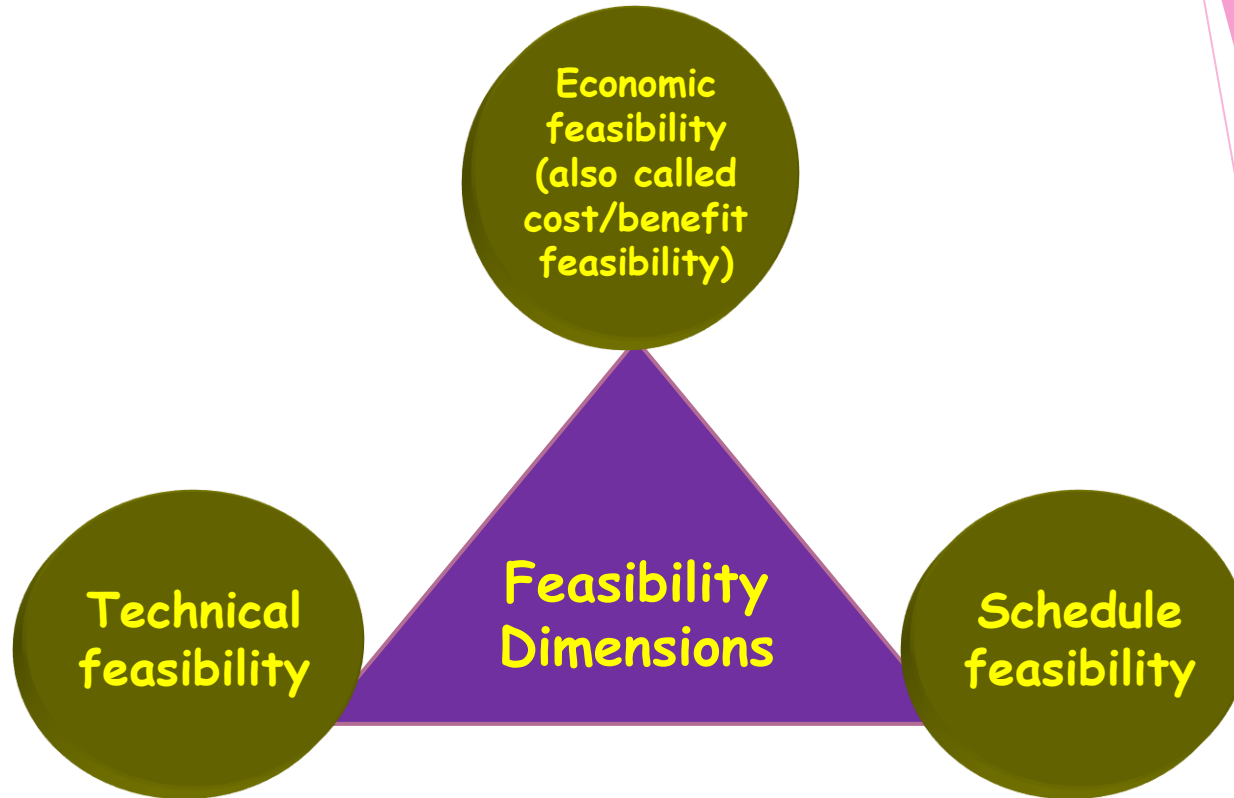
- Phases between feasibility study and testing
 - known as **development phases**.
- Among all life cycle phases
 - **maintenance phase consumes maximum effort.**
- Among development phases,
 - testing phase consumes the maximum effort.



Classical Waterfall Model (CONT.)

- ▶ **Most organizations usually define:**
 - ▶ entry and exit criteria for every phase.

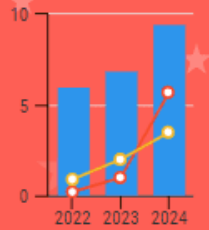
Feasibility Study



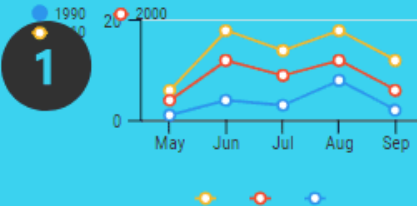
Feasibility Study

- ▶ Main aim of feasibility study: **determine whether developing the product is**
 - ▶ financially worthwhile
 - ▶ technically feasible.

5 Types of Feasibility Studies



1



1. Legal Feasibility

Performed to understand if the proposed plan conform the legal and ethical requirements.

2

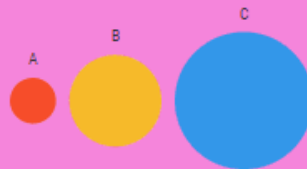


2019 2020 2021

2. Economic Feasibility

Involves a cost benefits analysis to identify how well, or how poorly, a project will be concluded.

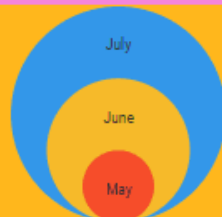
3



3. Technical Feasibility

Process of validating the technical resources and capabilities to convert the ideas into working systems.

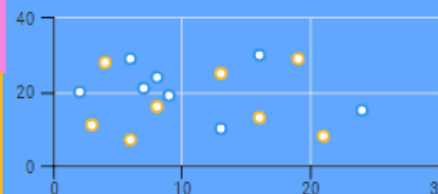
5



Scheduling Feasibility

Measure of how reasonable the project duration is.

4



1990 2000 2010

4. Operational Feasibility

Performed to understand well a proposed system solves the problems.

First roughly understand **what the customer wants**:

- different data which would be input to the system,
- processing needed on these data,
- output data to be produced by the system,
- various constraints on the behavior of the system.



Activities during Feasibility Study

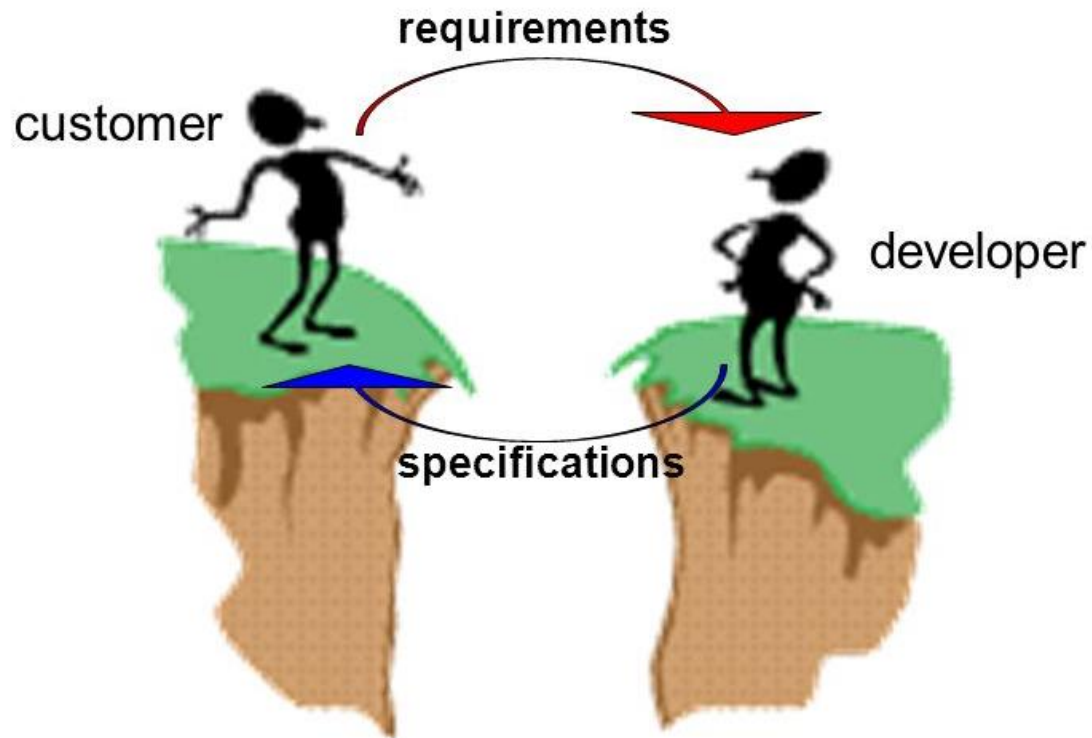
- ▶ Work out an overall understanding of the problem.
- ▶ Formulate different solution strategies.
- ▶ Examine alternate solution strategies in terms of:
 - ▶ resources required,
 - ▶ cost of development, and
 - ▶ development time.

Activities during Feasibility Study

- ▶ Perform a cost/benefit analysis:
 - ▶ to determine which solution is the best.
- ▶ **you may determine that none of the solutions is feasible due to:**
 - ▶ high cost,
 - ▶ resource constraints
 - ▶ technical reasons.



Requirements Analysis and Specification

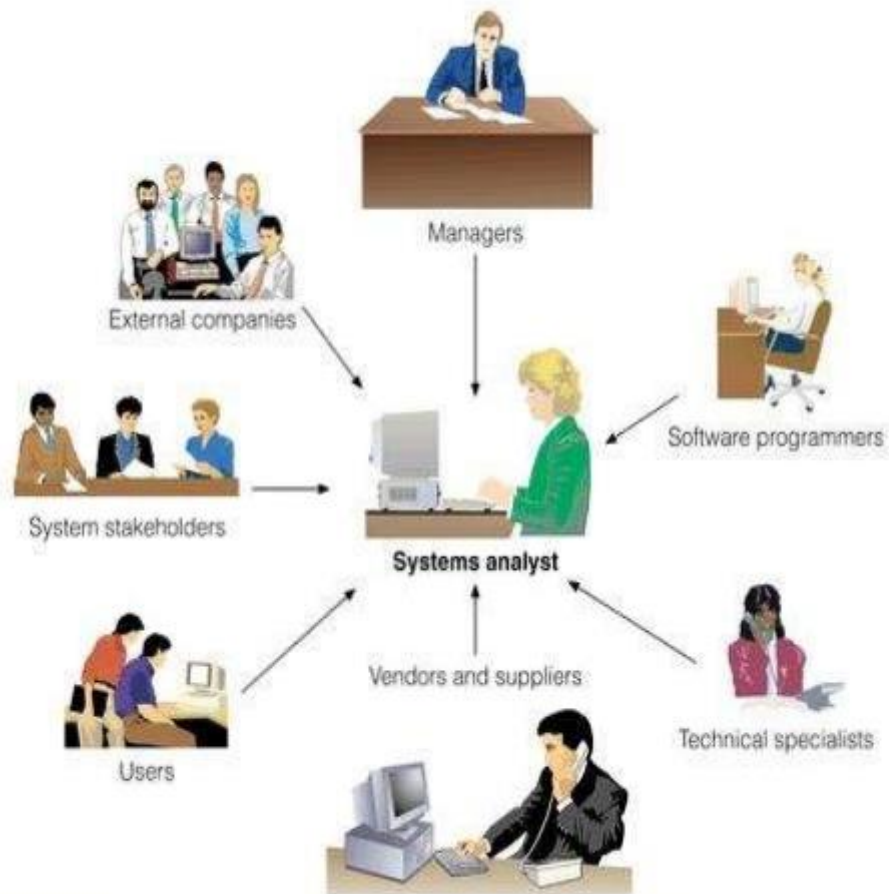


Requirements Analysis and Specification

- ▶ **Aim of this phase:**
 - ▶ understand the **exact requirements** of the customer,
 - ▶ document them properly.
- ▶ **Consists of two distinct activities:**
 - ▶ requirements gathering and analysis
 - ▶ requirements specification.

Goals of Requirements Analysis

- ▶ **Collect all related data from the customer:**
 - ▶ **analyze the collected data to clearly understand what the customer wants,**
 - ▶ **find out any inconsistencies and incompleteness in the requirements,**
 - ▶ **resolve all inconsistencies and incompleteness.**



Requirements Gathering

- ▶ **Gathering relevant data:**
 - ▶ usually collected from the end-users through interviews and discussions.
 - ▶ For example, for a business accounting software:
 - ▶ interview all the accountants of the organization to find out their requirements.

Requirements Analysis (CONT.)

- ▶ **The data you initially collect from the users:**
 - ▶ **would usually contain several contradictions and ambiguities:**
 - ▶ **each user typically has only a partial and incomplete view of the system.**

Requirements Analysis (CONT.)

- ▶ **Ambiguities and contradictions:**
 - ▶ must be identified
 - ▶ resolved by discussions with the customers.
- ▶ **Next, requirements are organized:**
 - ▶ into a **Software Requirements Specification (SRS)** document.

Requirements Analysis (CONT.)

- ▶ Engineers doing requirements analysis and specification:
 - ▶ are designated as analysts.

Design

- ▶ **Design phase transforms requirements specification:**
 - ▶ into a form suitable for implementation in some programming language.

Design

- ▶ In technical terms:
 - ▶ during design phase, software architecture is derived from the SRS document.
- ▶ Two design approaches:
 - ▶ traditional approach,
 - ▶ object oriented approach.

Traditional Design Approach

- ▶ Consists of two activities:
 - ▶ Structured analysis
 - ▶ Structured design

Structured Analysis Activity

- ▶ Identify all the functions to be performed.
- ▶ Identify data flow among the functions.
- ▶ Decompose each function recursively into sub-functions.
 - ▶ Identify data flow among the subfunctions as well.

Structured Analysis (CONT.)

- ▶ Carried out using **Data flow diagrams (DFDs)**.
- ▶ After structured analysis, carry out structured design:
 - ▶ architectural design (or high-level design)
 - ▶ detailed design (or low-level design).

Structured Design

- ▶ High-level design:
 - ▶ decompose the system into modules,
 - ▶ represent relationships among the modules.
- ▶ Detailed design:
 - ▶ different modules designed in greater detail:
 - ▶ data structures and algorithms for each module are designed.

Object Oriented Design

- ▶ First identify various objects (real world entities) occurring in the problem:
 - ▶ identify the relationships among the objects.
 - ▶ For example, the objects in a pay-roll software may be:
 - ▶ employees,
 - ▶ managers,
 - ▶ pay-roll register,
 - ▶ Departments, etc.

Object Oriented Design (CONT.)

- ▶ **Object structure**
 - ▶ further refined to obtain the detailed design.
- ▶ **OOD has several advantages:**
 - ▶ lower development effort,
 - ▶ lower development time,
 - ▶ better maintainability.

Implementation

- ▶ Purpose of implementation phase (**coding** phase):
 - ▶ translate software design into source code.

Implementation

- ▶ **During the implementation phase:**
 - ▶ each module of the design is coded,
 - ▶ each module is unit tested
 - ▶ tested independently as a stand alone unit, and debugged,
 - ▶ each module is documented.

Implementation (CONT.)

- ▶ **The purpose of unit testing:**
 - ▶ test if individual modules work correctly.
- ▶ **The end product of implementation phase:**
 - ▶ a set of program modules that have been tested individually.

Integration and System Testing

- ▶ Different modules are integrated in a planned manner:
 - ▶ **modules are almost never integrated in one shot.**
 - ▶ Normally integration is carried out through a number of steps.
- ▶ During each integration step,
 - ▶ the partially integrated system is tested.

Integration and System Testing

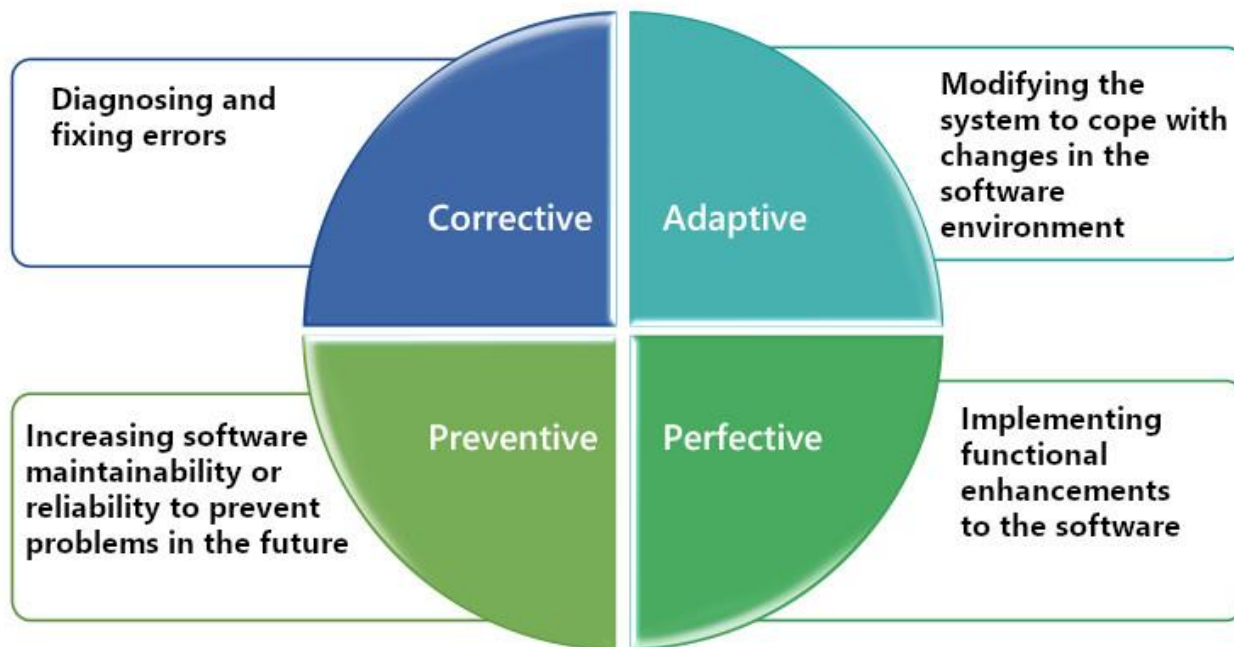


System Testing

- ▶ After all the modules have been successfully integrated and tested:
 - ▶ **system testing is carried out.**
- ▶ Goal of system testing:
 - ▶ ensure that the developed system functions according to its requirements as specified in the SRS document.

Maintenance

- ▶ **Maintenance of any software product:**
 - ▶ requires much more effort than the effort to develop the product itself.
 - ▶ development effort to maintenance effort is typically 40:60.



Maintenance (CONT.)

▶ Corrective maintenance:

- ▶ Correct errors which were not discovered during the product development phases.

▶ Perfective maintenance:

- ▶ Improve implementation of the system
- ▶ enhance functionalities of the system.

▶ Adaptive maintenance:

- ▶ Port software to a new environment,
 - ▶ e.g. to a new computer or to a new operating system.